

WATERWALKER · REQUIREMENTS

WaterWalker requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwalker		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3. Record Amish's decisions (WWK-DDR-001); redefine R1 (four jerrycans or two clay pots) and R11 (budget covers the unassisted first prototype with mounting points); add R13 (assist-ready to SwapCell interface v0.3); add status from WWK-CAL-001
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

WaterWalker requirements

Four of the thirteen requirements are not met on paper (R2, R3, R4 and R9) and six are at risk, according to the TRL 3 calculations in WWK-CAL-001. Targets are still desk proposals and must be revised from co-design sessions with the intended users (WWK-PRB-001) before the design is frozen. On 2026-09-25 Amish decided to build the first prototype without assist but with mounting points, keep the \$450 budget, and carry four jerrycans or two clay pots (WWK-DDR-001); R1 and R11 are redefined to match, and R13 is new.

Table 1. Requirements and status at TRL 3 (WWK-CAL-001).

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
R1	Carry a household's water in one trip	80 L of water in four 20 L jerrycans, or 40 L in two 20 L clay pots; rated gross payload 90 kg including containers (redefined 2026-09-25, WWK-DDR-001)	Cradle layout in the parametric model; later load test	At risk: two 380 mm pots fit with 8 mm to spare
R2	Easy to push on firm ground	Sustained push force 60 N or less at rated load on a firm, level dirt path	Rolling resistance calculation; later force gauge on a field path	Not met: 24.6 to 61.4 N
R3	Usable on loose sand	Sustained push force 150 N or less at rated load on loose sand, with assist if fitted	Calculation from sinkage and rolling resistance; later field test	Not met: 242 to 368 N (no assist on the first prototype)
R4	Climb and descend slopes	Climb a 10 % grade on firm ground with a push force of 180 N or less for up to 100 m; descend a 10 % grade under control with the service brake	Force calculation; brake sizing	Not met: 183.4 N at the rough end; descent met on paper
R5	Turn without lifting	180 degree turn within a 4.0 m diameter circle without lifting or skidding a wheel; reverse by pulling back	Geometry of the parametric model	Met: 3.66 m
R6	Fit narrow paths and gates	Overall width 900 mm or less	Parametric model	At risk: 898 mm
R7	Walk-in fit for all users	No step-over at the rear entry (50 mm or less); clear walking width 600 mm or more; hip bar height adjustable 850 to 1,050 mm; skirt guards on both rear wheels	Parametric model; later co-design fitting sessions	At risk: walking width 610 mm
R8	Easy loading	Containers lifted no higher than 450 mm above the ground; loadable from either side without removing parts	Parametric model	Met: 364 mm
R9	Light enough to handle empty	Empty mass 35 kg or less without assist, 42 kg or less with assist	Mass roll-up from the parametric model	Not met: 40.8 kg (47.3 kg)
R10	Safe braking and parking	Service brake on both rear wheels operable while walking; parking brake holds the rated gross mass on a 20 % grade	Brake force calculation	At risk: 189 N at the latch; tire friction 0.41 needed
R11	Affordable first prototype	Parts for the first prototype, without assist and with assist mounting points, \$450 or less (<code>project.yaml</code> budget, kept 2026-09-25). The budget excludes the optional assist kit, which is a later second prototype, and the SwapCell pack,	Priced BOM	At risk: \$430

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
		which is priced once in the SwapCell BOM (redefined 2026-09-25, WWK-DDR-001)		
R12	Repairable in a rural market	All wear parts (tires, tubes, spokes, bearings, cables, brake shoes) are standard 26 in bicycle parts; mild steel frame repairable by a local welder; only common hand and bicycle tools needed	Design review of the BOM; later review with a local mechanic	At risk: 100 mm drum hubs are less common than rim-brake hubs
R13	Assist-ready (new 2026-09-25)	The first prototype carries mounting points for a later assist kit built to SwapCell interface v0.3: a plate for a latch class V1 vehicle receiver above the tires, with the pack's back and lid faces open to air; rear dropouts that accept a 100 mm hub motor with a torque-arm tab; a boss for a push sensor at the hip bar. The receiver fits a 10 kΩ INTERLOCK coding resistor (item W), and the assist draws no more than the 15 A legacy-mode limit	Parametric model; later latch class V1 test	Not verifiable at TRL 3: geometry present, 7.6 A peak

■ Assumptions

- A 20 L jerrycan weighs about 1.1 kg empty and measures about 360 x 175 x 430 mm. A 20 L clay pot is assumed to be about 380 mm in diameter and 5 to 8 kg empty.
- Sustained push forces are set with reference to published manual-handling guidance, which suggests roughly 100 to 150 N for sustained pushing by women over long distances and more for short efforts. The limits must be confirmed with the users, including girls and older women.
- A household of five uses about 100 L per day (about 20 L per person).
- The first prototype has no assist (WWK-DDR-001), so R3 cannot be met by it; R3 stays as the target for the later assist prototype, which needs about 240 N of thrust (two hub motors) to meet it on the loosest sand (WWK-CAL-001).
- R5 allows a larger circle than a compact rollator because the concept needs a long wheelbase to keep the swivelling front wheels clear of the cradle (WWK-PRC-001). A three-point turn is expected on paths narrower than 4 m.